

ALFAAL03361

## 1-Undecanol

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | 1-Undecanol<br>1-Undecanol                                                                                                                                                                                                                                         |
| Cat No. :                     | L03361                                                                                                                                                                                                                                                             |
| Synonyms                      | Undecyl alcohol                                                                                                                                                                                                                                                    |
| CAS No                        | 112-42-5                                                                                                                                                                                                                                                           |
| Molecular Formula             | C11 H24 O                                                                                                                                                                                                                                                          |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                           |                            |                        |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------|
| Physical State<br>Liquid                                                                                                                  | Appearance<br>Light yellow | Odor<br>Characteristic |
| <b>Emergency Overview</b><br>May be harmful if swallowed. Causes serious eye irritation. Toxic to aquatic life with long lasting effects. |                            |                        |

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 5 |
| Serious Eye Damage/Eye Irritation | Category 2 |
| Chronic aquatic toxicity          | Category 2 |

#### Label Elements



Signal Word

Warning

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**Hazard Statements**

H303 - May be harmful if swallowed

H319 - Causes serious eye irritation

H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear eye protection/ face protection

**Response**

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May be harmful if swallowed. Causes serious eye irritation.

**Environmental hazards**

Toxic to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil. The product is insoluble and floats on water.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component   | CAS No   | Weight % |
|-------------|----------|----------|
| 1-Undecanol | 112-42-5 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of

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contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems,

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should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                 |                          |                                          |
|---------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>               | Light yellow             |                                          |
| <b>Physical State</b>           | Liquid                   |                                          |
| <b>Odor</b>                     | Characteristic           |                                          |
| <b>Odor Threshold</b>           | No data available        |                                          |
| <b>pH</b>                       | No information available |                                          |
| <b>Melting Point/Range</b>      | 11 °C / 51.8 °F          |                                          |
| <b>Softening Point</b>          | No data available        |                                          |
| <b>Boiling Point/Range</b>      | 146 °C / 294.8 °F        | @ 30 mmHg                                |
| <b>Flash Point</b>              | > 112 °C / > 233.6 °F    | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | No data available        |                                          |
| <b>Flammability (solid,gas)</b> | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>         | No data available        |                                          |
| <b>Vapor Pressure</b>           | No data available        |                                          |
| <b>Vapor Density</b>            | 5.9                      | (Air = 1.0)                              |

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|                                         |                          |        |
|-----------------------------------------|--------------------------|--------|
| Specific Gravity / Density              | 0.831                    |        |
| Bulk Density                            | Not applicable           | Liquid |
| Water Solubility                        | Immiscible               |        |
| Solubility in other solvents            | No information available |        |
| Partition Coefficient (n-octanol/water) |                          |        |
| Component                               | log Pow                  |        |
| 1-Undecanol                             | 4.28                     |        |
| Autoignition Temperature                | No data available        |        |
| Decomposition Temperature               | No data available        |        |
| Viscosity                               | No data available        |        |
| Explosive Properties                    | No information available |        |
| Oxidizing Properties                    | No information available |        |

|                   |           |
|-------------------|-----------|
| Molecular Formula | C11 H24 O |
| Molecular Weight  | 172.31    |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                 |
|--------------------------|---------------------------------|
| Stability                | Stable under normal conditions. |
| Hazardous Reactions      | None under normal processing.   |
| Hazardous Polymerization | No information available.       |
| Conditions to Avoid      | None known.                     |
| Materials to avoid       | Strong acids. Oxidizing agent.  |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component   | LD50 Oral             | LD50 Dermal              | LC50 Inhalation                          |
|-------------|-----------------------|--------------------------|------------------------------------------|
| 1-Undecanol | LD50 = 3 g/kg ( Rat ) | LD50 > 5 g/kg ( Rabbit ) | LC50 > 700 mg/m <sup>3</sup> ( Rat ) 6 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

## (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

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|                                            |                                                                |
|--------------------------------------------|----------------------------------------------------------------|
| (h) STOT-single exposure;                  | No data available                                              |
| (i) STOT-repeated exposure;                | No data available                                              |
| Target Organs                              | None known.                                                    |
| (j) aspiration hazard;                     | No data available                                              |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated. |
| Symptoms / effects, both acute and delayed | No information available                                       |

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component   | Freshwater Fish                                           | Water Flea | Freshwater Algae | Microtox |
|-------------|-----------------------------------------------------------|------------|------------------|----------|
| 1-Undecanol | LC50: = 1.04 mg/L, 96h flow-through (Pimephales promelas) |            |                  |          |

**Persistence and Degradability****Persistence****Degradation in sewage treatment plant**

May persist.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Product has a high potential to bioconcentrate

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| 1-Undecanol | 4.28    | 2280                          |

**Mobility in soil**

Spillage unlikely to penetrate soil The product is insoluble and floats on water Is not likely mobile in the environment due its low water solubility Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

**Endocrine Disruptor Information****Persistent Organic Pollutant****Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## SECTION 14. TRANSPORT INFORMATION

## 1-Undecanol

**Road and Rail Transport**

UN-No UN3082  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
Technical Shipping Name 1-Undecanol  
Hazard Class 9  
Packing Group III

**IMDG/IMO**

UN-No UN3082  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
Technical Shipping Name 1-Undecanol  
Hazard Class 9  
Packing Group III

**IATA**

UN-No UN3082  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
Technical Shipping Name 1-Undecanol  
Hazard Class 9  
Packing Group III

Special Precautions for User No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component   | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| 1-Undecanol | -                                                   | -                                       | X    | X     | 203-970-5 | X    | X   | X     | X    | X    | X    | KE-35068 |

**National Regulations****SECTION 16. OTHER INFORMATION**

Prepared By Health, Safety and Environmental Department  
Creation Date 05-Sep-2014  
Revision Date 27-Apr-2024  
Revision Summary New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

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Legend**CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**